Scope of Work: Project #3 – Transportation Data Analysis and Simulation

Client: Federal Highway Administration

From: Macy Dubes

Date: March 31, 2026

Project Goals:

The main goal of this project is to transition from data observation to transportation modeling. By using Python analysis, I will find core trends in the National Household Travel Survey (NHTS) and validate the Intelligent Driver Model (IDM) against real-world NGSIM vehicle trajectories. This project aims to provide the client with the framework for predicting driver behavior.

Project Tasks:

To achieve the project goals, the following technical tasks will be completed:

- Task 1: Data Setup & Cleaning: Start the Python workspace using pandas, numpy, and seaborn. Import the raw NHTS and NGSIM datasets and run the data using .head and .tail to identify missing values.
- Task 2: Vehicle Characteristic Visualization (NHTS): Develop three distinct statistical visualizations to summarize the US vehicle fleet:
    - A Bar chart to categorize the frequency of vehicle types.
    - A Histogram to determine the distribution of vehicle ages.
    - A Boxplot to compare vehicle age spreads across different fuel categories

- Task 3: Time-Series Behavior Analysis (NGSIM): Pick specific vehicle trajectory pairs (Leader vs. Follower) to visualize real-world driving behavior. This includes plotting speed profiles and calculating gap distance between vehicles.
- Task 4: IDM Simulation: Make a Python function representing the Intelligent Driver Model (IDM). Using Euler's Method for numerical integration, the team will simulate a follower vehicle's response to a real-world leader's movements, updating position and speed at 0.1-second intervals.
- Task 5: Comparative Validation: Generate a final comparison plot by putting the simulated IDM acceleration against the actual recorded NGSIM acceleration to evaluate the model's accuracy and reliability.

Project Deliverables:

Deliverable	Description	Due Date
Python Analysis Script	A functioning .ipynb or .py file containing all visualization and simulation code.	March 31, 2026
Annotated Code Document	A pdf explaining the logic behind the Python script.	March 31, 2026
Final Technical Report	A report summarizing findings from the NHTS data and the IDM simulation results.	March 31, 2026
GitHub Repository	A repository containing all project files, including a README.	March 31, 2026
Project Timesheet/Gantt	A record of hours spent on data cleaning, coding, and reporting.	March 31, 2026